Fetch and supply limitation

Simon Frost

Table of contents

1 Overview 1

2 Fetch with unlimited supply 2

3 Supply limitation 3

References 5

1 Overview

AeoLiS solves an advection equation for the sediment concentration in the air `C`, which relaxes towards the saturated concentration `Cu` over the adaptation time `T` (Hoonhout and de Vries 2016):

$$\frac{\partial C}{\partial t} + u\frac{\partial C}{\partial x} = \frac{C_u - C}{T}.$$

In steady state with unlimited supply, the concentration grows along the fetch as the solution of an ordinary differential equation, `dC/dx = (Cu - C) / (u T)`. This vignette checks the model against that solution, computed with OrdinaryDiffEq, and then limits the supply with a non-erodible layer.

```
ENV["GKSwstype"] = "100"
using Aeolis
using OrdinaryDiffEq
using Plots
```

2 Fetch with unlimited supply

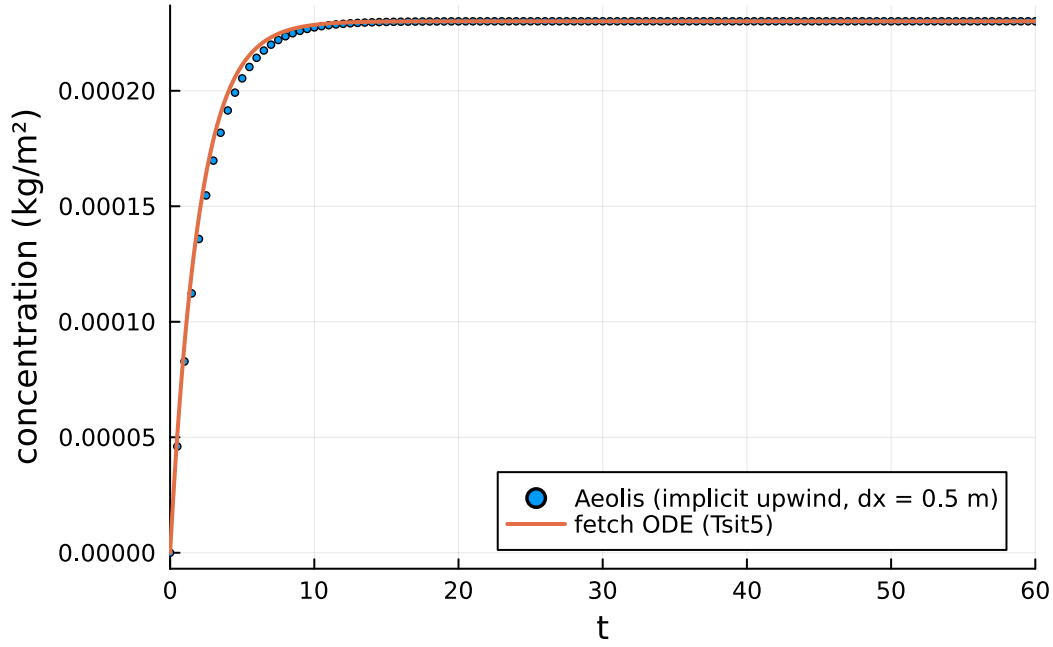
A flat transect of 60 m with grain velocities equal to the wind (`method_grainspeed = :windspeed`) makes `u` and `Cu` uniform. The adaptation time is raised to 0.2 s so the adaptation length `u * T` spans a few cells.

```
x = collect(0.0:0.5:60.0)
p = AeolisParameters(; xgrid_file=x, bed_file=fill(2.0, length(x)),
                       wind_file=[0.0 10.0 270.0; 86400.0 10.0 270.0],
grain_size=0.0003,
                       method_grainspeed=:windspeed, T=0.2, dt=60.0, tstop=600.0,
                       output_times=600.0, output_vars=[:Ct, :Cu, :u])
sol = solve(AeolisProblem(p))
C = transect(sol, :Ct)[:, 1, end]
Cu = transect(sol, :Cu)[2, 1, end]
u = transect(sol, :u)[2, 1, end]
(Cu=Cu, u=u, adaptation_length=u * p.T)
```

```
(Cu = 0.00023007629540473118, u = 10.0, adaptation_length = 2.0)
```

The fetch equation, solved with an adaptive ODE solver:

```
fetch(C, par, x) = (par.Cu - C) / (par.u * par.T)
ode = ODEProblem(fetch, 0.0, (0.0, 60.0), (; Cu, u, T=p.T))
exact = solve(ode, Tsit5(); reltol=1e-10, abstol=1e-12)
scatter(x, C; ms=2, label="Aeolis (implicit upwind, dx = 0.5 m)", xlims=(0, 30),
        xlabel="fetch (m)", ylabel="concentration (kg/m²)")
plot!(exact; lw=2, label="fetch ODE (Tsit5)")
```



The implicit first-order upwind scheme of AeoLiS gives the exponential approach with a numerical diffusion that depends on the cell size `dx` compared with `u T`. For each cell the discrete solution is `C_i = (C_{i-1} + a Cu) / (1 + a)` with `a = dx / (u T)`, so it lags the exact solution; the gap shrinks as `dx / (u T)` falls:

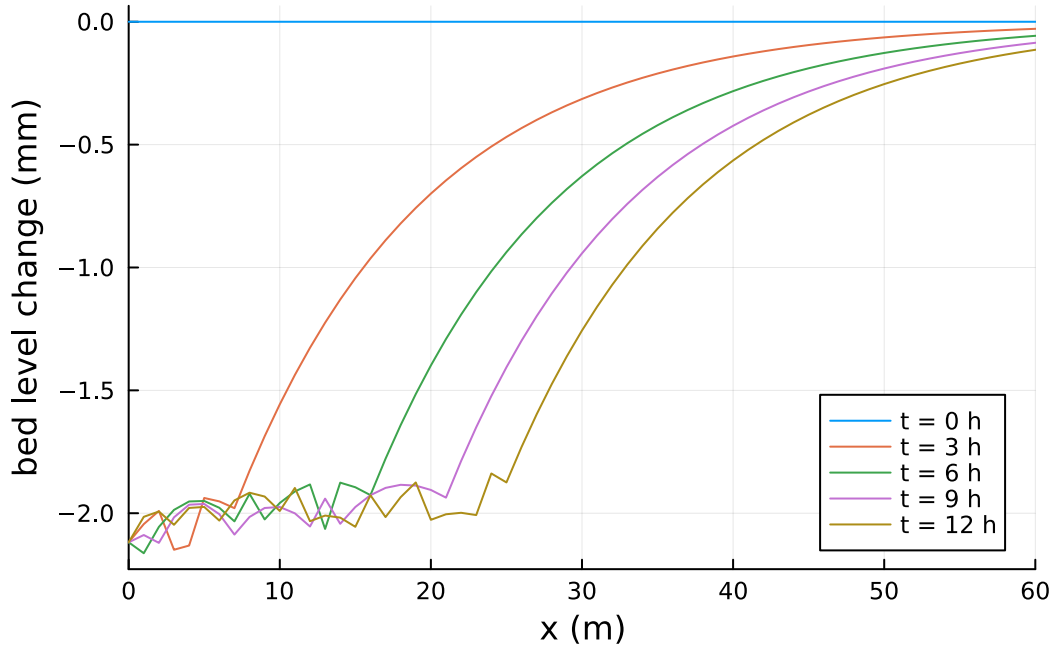
```
err(dx) = begin
    xx = collect(0.0:dx:60.0)
    q = remake_parameters(p; xgrid_file=xx, bed_file=fill(2.0, length(xx)))
    s = solve(AeolisProblem(q))
    c = transect(s, :Ct)[:, 1, end]
    maximum(abs.(c .- exact.(xx))) / Cu
end
[(dx=dx, max_relative_error=round(err(dx); sigdigits=3)) for dx in (2.0, 1.0, 0.5,
0.25)]
```

```
4-element Vector{@NamedTuple{dx::Float64, max_relative_error::Float64}}:
 (dx = 2.0, max_relative_error = 0.132)
 (dx = 1.0, max_relative_error = 0.0766)
 (dx = 0.5, max_relative_error = 0.0417)
 (dx = 0.25, max_relative_error = 0.0219)
```

3 Supply limitation

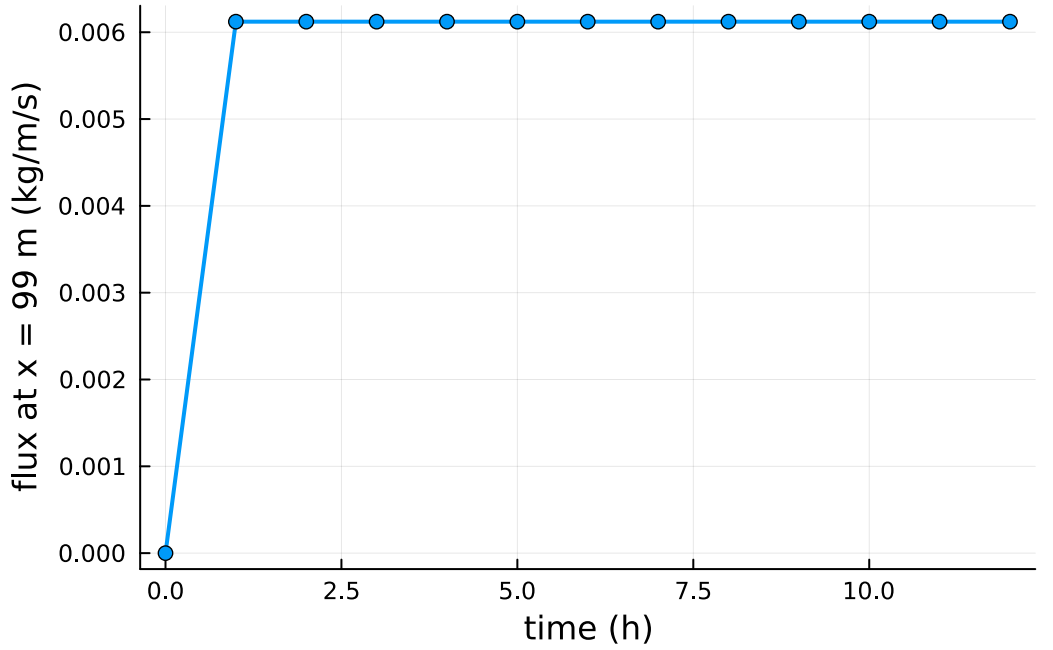
Now the bed carries only 2 mm of loose sand over a non-erodible layer (`ne_file`, with `th_nelayer` making the threshold infinite where the bed reaches it). The upwind part of the transect is stripped of sand, after which the fetch starts further downwind.

```
x = collect(0.0:1.0:100.0)
q = AeolisParameters(; xgrid_file=x, bed_file=fill(2.0, length(x)),
                     ne_file=fill(2.0 - 0.012, length(x)), th_nelayer=true,
                     wind_file=[0.0 12.0 270.0; 86400.0 12.0 270.0],
grain_size=0.0003,
                     nlayers=3, layer_thickness=0.002, dt=600.0, tstop=12 * 3600.0,
                     output_times=3600.0, output_vars=[:zb, :Ct, :qs])
sl = solve(AeolisProblem(q))
zb = transect(sl, :zb)
plot(x, 1000 .* (zb[:, 1:3:end] .- zb[:, 1]); xlabel="x (m)",
     ylabel="bed level change (mm)", label=permutedims(["t = $(Int(t / 3600)) h"
                                                        for t in sl.t[1:3:end]]),
     xlims=(0, 60))
```



The flux leaving the transect falls as the sand runs out:

```
qs = transect(sl, :qs)
plot(sl.t ./ 3600, qs[end - 1, 1, :]; lw=2, marker=:circle, xlabel="time (h)",
     ylabel="flux at x = 99 m (kg/m/s)", label="")
```



References

Hoonhout, Bas M., and Sierd de Vries. 2016. "A Process-Based Model for Aeolian Sediment Transport and Spatiotemporal Varying Sediment Availability." *Journal of Geophysical Research: Earth Surface* 121: 1555–75. https://doi.org/10.1002/2015JF003692.